Arthur **Fournier**

Machine Learning · Robotics · Computer Vision

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Education

Mines Paris – PSL — *Paris, France*

Double degree with École polytechnique · Robotics track — *2026 – 2027 (expected)*

- Nonlinear control of manipulators and mobile robots, reinforcement learning for robotics, estimation and filtering, 3D vision.

Master MVA – ENS Paris-Saclay — *Paris, France*

Auditing student (non-degree) · 3D computer vision, robotics, point clouds, 3D modeling and optimal transport — *2026 – Present*

École polytechnique — *Palaiseau, France*

MSc in Engineering (Ingénieur Polytechnicien program) — *2023 – 2027 (expected)*

- Machine learning, deep learning, graph learning, reinforcement learning, generative AI, computer vision, statistics, optimization and control.

Lycée Hoche — *Versailles, France*

Preparatory classes in Mathematics, Physics and Computer Science (MPI) — *2021 – 2023*

Experience

National University of Singapore & CNRS (IPAL joint lab) — *Singapore*

Research Intern, Event-based Vision for Assistive Robots — *Mar 2026 – Jul 2026*

- Cut the trajectory error of DEVO, the state-of-the-art learned monocular event-camera odometry, by 41% on the EVIMO2 dataset with a ConvLSTM that learns to ignore independently moving objects.
- Showed that training it jointly with the odometry beats training it separately, and that a label-free, self-supervised variant still cuts error by 37%.
- Beat vanilla DEVO in 15 of 20 zero-shot evaluations on 4 unseen datasets; paper and code release in progress.

Headmind Partners AI — *Paris, France*

AI Engineer Intern — *Jun 2025 – Sep 2025*

- Designed and deployed a multi-agent LLM pipeline where role-played HR, CTO and CEO agents assess each CV, then rank candidates.
- Took it from prototype to a tested web app (OpenAI API, Hugging Face embeddings, Streamlit), used internally on 50+ real applications.

Projects

Reinforcement Learning for the Block Blast Puzzle Game — *GitHub*

PyTorch, Gymnasium, Weights & Biases — *Feb 2026 – Mar 2026*

- With 4 teammates, trained a reinforcement learning agent to play Block Blast: after designing the environment, tested DQN, Rainbow and an afterstate value network, and kept the latter, which survives 209 moves on average (vs. 44 for a greedy baseline).

edge-mppi: Real-time GPU Model Predictive Control *(in progress)* — *GitHub*

CUDA, C++, Python — *Sep 2026 – Present*

- Modeled a car-like vehicle (dynamic bicycle model) and implemented an MPPI controller driving full laps.
- Now porting it to a fused CUDA kernel to run 8,192 rollouts within 20 ms.

Makeitalive: Animating a Single Landscape Photo — *GitHub*

PyTorch, Diffusers — *Feb 2026 – Mar 2026*

- With a teammate, iterated from an optical-flow U-Net to LoRA adapters to full fine-tuning of Stable Video Diffusion's temporal layers on drone footage, turning a single photo into a video with cinematic drone-like parallax.

Retrieval Assistant for Emergency Call Handlers — *GitHub*

Python, FAISS, Llama — *Sep 2024 – May 2025*

- In a student team, built an on-premise RAG assistant over a medical regulation guide, giving source-linked answers, and designed its integration into a SAMU call center's workflow through interviews and live-call observation with operators, then tested it there.

X-Forum Careers Fair Platform — *xforum.binets.fr*

React, TypeScript, Django, Docker, CI/CD — *Apr 2026 – Present*

- Co-built and deployed the portal of École polytechnique's careers fair, serving 250 companies and 3,000 students and processing 160 stand orders, with passwordless login and DocuSign-signed purchase orders.

Skills

Programming	Python, C, C++, CUDA, Java, SQL, TypeScript
Machine Learning	PyTorch, Hugging Face (Transformers, Diffusers), scikit-learn, pandas
Systems & Tools	Pinocchio, OpenCV, Git, Docker, CI/CD, Slurm, Weights & Biases
Languages	French (native), English (C1), Spanish (B2)
Leadership	Gendarmerie officer cadet, aide to the commander of a 250-strong company (2023–24) · X-Forum logistics team
Interests	Basketball (university team), BMX racing (10 years, inter-regional), piano and guitar